

Answers and questions for the advisors

Giovanni Mantovani — 7 August 2026. Companion to [paper/main.pdf](#) and [compendium/main.pdf](#).

Every answer below separates four things explicitly: what is an **established result**, what is **numerical evidence**, what is **current interpretation**, and what is **unresolved**. Where the honest answer is "partially", it says so.

Part I — Questions posed in the meetings

A. Can belief propagation for continuous non-Gaussian variables be made operational?

Established result. Yes, at the level of functional messages. Coordinatewise OU noising contributes one unary factor per site, so the posterior factor graph of a Markov chain is again that chain (proved in [compendium Ch. 6](#)). A chain is a tree, so sum-product returns the exact single-site and pairwise marginals. Nothing here is approximate.

Numerical evidence. Making it *computable* requires representing the messages. A truncated uniform grid with N_g points and trapezoidal weights turns each message update into one $N_g \times N_g$ matrix–vector product, so a sequence costs $O(nN_g^2)$ against $O(N_g^n)$ for naive marginalisation — at $n = 32$, $N_g = 401$, about 5×10^6 operations. Batching over sequences makes each update a single GEMM, which is why the same code runs on a GPU without restructuring.

Accuracy then depends on truncation and quadrature, and these are now measured separately (Fig. 2 of the note). Truncation responds to the domain A and not to N_g : worst boundary mass over 32 sites falls $3.4 \times 10^{-4} \rightarrow 1.4 \times 10^{-8} \rightarrow 1.4 \times 10^{-10}$ at $A = 4, 8, 10$. Quadrature responds to the spacing: the interior column-mass residual follows $O(h^2)$ exactly, a factor of four per halving. Validation against the closed-form Gaussian posterior mean gives 9.2×10^{-15} ; against brute-force enumeration of the discretised posterior, 1.6×10^{-14} .

Interpretation. "Operational" holds for a scalar state. The $O(N_g^2)$ per update is a dense matrix product; for a translation-invariant kernel $K(a'|a) = \varphi(a' - \rho a)$ a convolutional structure is available and unexploited, which would bring it to $O(N_g \log N_g)$.

Unresolved. Nothing here scales to a vector-valued state without a further approximation: the grid cost is exponential in the per-site dimension.

B. Is a Gaussian-message approximation justified, and how inaccurate is it?

Established result. On this model, single-Gaussian message closure computes exactly the **LMMSE estimator of the covariance-matched Gaussian model** (Proposition 1 of the note; derivation in [compendium Ch. 8](#)). Both moment maps read only the mean and variance of the innovation, so the recursion returns the same function of x for every innovation law with moments $(0, q)$ — in particular the same as on the Gaussian chain, where it is exact. It is therefore exact for the Gaussian state-space model and, for every other member of the family, exactly the best *linear* estimator.

A terminological point we would like to confirm. We deliberately do **not** call this AMP. A degree-two chain supplies no large-degree central-limit justification for that terminology, so the result is proved directly rather than argued asymptotically. We now use one name throughout, **Gaussian second-order baseline**, and note once that on this linear model message closure and outright model replacement coincide (they separate for richer message families).

Numerical evidence. The gap is largest at *small* t and decays monotonically. On a Laplace chain at $\rho = 0.85$ the median relative score error falls from 0.175 at $t = 0.08$ to 7.6×10^{-5} at $t = 2.4$ (Fig. 3, from `outputs/exp_02_laplace_gaussian_message_error/laplace_summary.csv`). The ordering across families follows innovation excess kurtosis. Large t Gaussianises the posterior, so the shape of φ survives only where the likelihood is weak.

One correction: an earlier draft quoted $0.198 \rightarrow 9.9 \times 10^{-5}$ for this pair. Those numbers do not appear in any committed output and were presumably from a superseded run. The committed medians are $0.175 \rightarrow 7.6 \times 10^{-5}$.

Unresolved. The zero-mean hypothesis in Proposition 1 is load-bearing (with $\mathbb{E}[a] = 1.3$ the estimator departs by 0.65 in sup-norm), and the *fitted* mixture is not constrained to satisfy it, so the proposition applies to the true kernel and only approximately to the estimated one.

C. Does knowing the clean distribution is Markov reduce the score-learning problem?

Established result. Structurally, yes: the posterior chain is preserved under coordinatewise noising, so an unrestricted n -dimensional regression is replaced by (i) learning a *local* transition kernel and (ii) two sweeps of inference. The parameters live on the clean chain and the noise level lives only in the likelihood, so one fitted kernel is a denoiser at **every** noise level with no further fitting.

Numerical evidence. Twelve free parameters against 25,248, at a relative denoising error 9–14% lower, across seven doublings of the data. Against a locality-respecting convolution the margin is 2–4. Parameter recovery is clean: ρ from 0.3 to $[0.8497, 0.8507]$, RMSE falling at roughly $M^{-1/2}$.

Interpretation. The gain depends on correct specification and on the representational capacity of K_θ . Both caveats are measurable and we measured them: capacity is the subject of question F; misspecification is not addressed at all (see "Unresolved").

Unresolved. Every result assumes the Markov structure is *correct*. We have no measurement of graceful degradation under a non-Markov prior, and that is the single largest gap in the story.

D. Can the unknown Markov model be learned efficiently through EM-style methods?

Established result. Yes, and the structure is unusually favourable. The complete-data log-likelihood is a sum over edges, so its θ -dependent part depends on the data only through the expected transition mass $\Xi_{kl} = \sum_\mu \sum_i P(a_{i-1} = u_k, a_i = u_l \mid x^\mu)$ — the continuum analogue of Baum–Welch counts. The M-step never revisits the data. Fisher’s identity gives the exact gradient of the *marginal* log-likelihood from one sum-product pass, so BP is never differentiated through; verified against finite differences to $\sim 10^{-9}$.

Precision about the algorithm, which the previous draft got wrong. The E-step is exact for the discretised model, not for the continuous one. The M-step for the mixture kernel is **not** an exact maximisation: it runs four inner conditional-maximisation sweeps, so the algorithm is **generalised EM of ECM type**. Monotone ascent is preserved; the exact-maximiser property is not. Monotonicity is observed: the largest decrease across iterations is zero in all eight recorded runs.

Numerical evidence. Both ρ and the innovation density are estimated. The **initial law** μ is fixed at $\mathcal{N}(0, 1)$ — which is the true one, so nothing is misspecified, but it is not recovered

either. Fisher information per sequence for the innovation variance falls 142E between $t = 0.05$ and $t = 1.6$; for the correlation, 26E. Higher-order structure is far more fragile under the channel than second-order structure.

Unresolved. Identifiability is assumed, not proved (Assumption 1 of the note). Injectivity of Gaussian deconvolution gives $P_t \Rightarrow P_0$; kernel identifiability additionally needs a fixed or separately identifiable μ with full support, identifiability of the parametric family (which for finite mixtures holds only on the interior of the parameter set), and identification modulo label permutation.

E. How local is the score computation? Can the required inference be localised?

This is the question we answer least well, and we want to be blunt about it.

Established result. The computation is *composed* of local updates — each message update touches one edge — but the marginal at site i depends on the whole sequence: information propagates from both ends. There is no finite exact receptive field in general.

Numerical evidence. `exp_11` and `exp_12` measure how the error of a *local predictor* falls with its window radius, and the fitted slopes match a closed-form $\log q$ prediction to 0.8–8%. That is evidence about how much context a *denoiser* needs to reach a given accuracy.

Interpretation. It is a proxy, not a measurement of the score’s locality. The two differ: the first is about approximating the score with a restricted architecture, the second is about the score’s own dependence structure.

Unresolved. We have no direct statement. The two experiments that would give one — windowed inference with the truncation error measured, or the Jacobian of the score with respect to distant observations — have not been run. And `exp_12`’s receptive field was selected on the evaluation set, so even the proxy is currently exploratory.

F. Does better pointwise denoising imply better generation?

Established result. No, not automatically. Reverse integration accumulates errors unevenly across the schedule, and the marginal likelihood and the squared-error loss are both dominated by the bulk of the distribution while shape statistics are driven by the tail.

Numerical evidence, in two parts.

The disagreement is real and general. At $C = 4$ the estimator is an order of magnitude better pointwise and three times better on second-order structure, and yet the convolution reproduces the innovation law more faithfully (kurtosis 1.357 vs 0.897, target 1.910). This holds across all four non-Gaussian families at matched covariance, at $n \in \{32, 64, 128\}$, and identically to three decimals under $N_g \in \{401, 801, 1601\}$. On the **Gaussian** chain it vanishes (disagreement -0.028 , within noise of zero) — the negative control the capacity account predicts and an information account does not.

The capacity sweep now covers both axes. This was the open item; the run has returned.

C	MSE vs Bayes denoiser	generated excess kurtosis
2	0.002281	0.034
4	0.000510	0.812
8	0.000276	1.319
12	0.000249	1.363
16	0.000234	1.487

Pointwise error improves 9.7E while generated kurtosis climbs monotonically. **The two axes do not trade off** anywhere in the range tested; at $C = 16$ the estimator leads the CNN on both simultaneously.

Interpretation. The disagreement at small C is a property of a too-small innovation model rather than of the information surviving the channel, since enlarging the model removes it at no cost on the other axis.

Unresolved. This does not show that information is never binding: the kurtosis curve has not saturated at $C = 16$. And the joint experiment is complete in three coordinates of five — marginal likelihood versus C and runtime versus C are still missing.

Part II — Our questions for Jérôme

1. **Framing.** ρ is estimated (initialised at 0.3, recovered to 0.850), so we kept "transition-kernel estimation" in the title rather than downgrading to "innovation law". But the kernel is constrained to the linear form $K(a'|a) = \varphi(a' - \rho a)$. Is the stronger title defensible, or would you rather the restriction were in the title?
 1. **Should ρ and μ both be learned next?** The E-step already computes the site-one statistic and discards it, so learning μ is a small change. It would also cost us the sufficiency argument for Ξ as currently stated. Worth it?
 1. **Strict stationarity.** Should we replace $a_1 \sim \mathcal{N}(0, 1)$ with each family's invariant law? It needs a fixed-point computation per family. We expect it to change nothing, but right now that is a prediction rather than a measurement.
 1. **Where does the generation result belong?** It is currently a full section of the main text. Now that the capacity sweep resolves it on both axes, is it a headline result or an appendix ablation?
 1. **Which experiment first?** Our ranking is (a) capacity versus pointwise error — *done*; (b) validation-based model selection for the neural baselines; (c) a direct locality experiment; (d) strict-stationarity rerun; (e) richer neural comparison. Would you reorder?
 1. **Is the local-CNN comparison enough?** Should we add a structured autoregressive or mixture-density baseline — something that is *given* the sequential structure but has to learn the transition, which is the fairest comparison we can think of?
 1. **Which generative metric should lead?** We currently emphasise innovation excess kurtosis because it is a scalar non-Gaussianity knob with a closed-form target at $t_{\min}$. KL and the covariance-lag error are also reported. Is there one you would rather see first?
 1. **Diffusion contribution or structured-inference contribution?** The result is arguably more about what structure buys in inference than about diffusion specifically. How should this be framed for the thesis?
 1. **How central should the locality/receptive-field direction stay?** It is the question we answer least well (Part I.E) and the one with the most room.
 1. **What would make this publishable rather than internal?** Specifically: how much of the identifiability question needs proving, and how much non-Markov robustness needs measuring?
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Part III — Proposed weekend work plan

Priority 1 — done in this revision. Notation (K for the kernel, N_g for the grid); stationarity corrected to covariance stationarity; the four exactness tiers separated and applied; the reverse-SDE convention boxed and used without variation; the $t \rightarrow 0$ claim corrected; private lecture notes removed repo-wide and replaced with public sources; numbered references; appendices A–J; ten figures generated from committed data.

Priority 2 — measurement gaps. Capacity versus pointwise error (**done**, resolved on both axes); domain/grid convergence (**done**, truncation and quadrature separated); reverse-sampler convergence (**done** over 100–800 steps). Remaining: validation-based neural model selection; marginal likelihood and runtime versus C .

Priority 3 — presentation. All figures regenerated, both PDFs compiled, advisor package prepared. Done.

Priority 4 — whatever you recommend after reading. Held open deliberately.

Part IV — Evidence map

answer	note §	compendium	figure / table	script	out
A: BP operational	§3, App. C	Ch. 6–7	Fig. 1	src/bp_grid.py	—
A: grid cost & accuracy	§4.1, App. E	Ch. 9	Fig. 2	exp_01, exp_18 -parts boundary	exp
B: LMMSE closure	§4.3, App. D	Ch. 8	Prop. 1	src/bp_gaussian.py	—
B: closure error vs t	§4.3	Ch. 8	Fig. 3, Fig. 10	exp_02, exp_03	exp
C: sample efficiency	§7	Ch. 11	Tab. 1, Fig. 5	exp_07	rej
C: cost of the structure	§9	Ch. 11	Fig. 9	exp_07	exp
D: EM/GEM, monotonicity	§5, App. F	Ch. 10	Fig. 4	exp_18 -parts emtrace	exp
D: parameter recovery	§5	Ch. 10	Fig. 4	exp_06	exp
D: Fisher identity	§5	Ch. 10	—	exp_08	exp
D: information loss 142/26	§5	Ch. 10	—	exp_06	exp
E: locality proxy	§9	Ch. 11	—	exp_11, exp_12	exp
F: disagreement	§8	Ch. 11	Tab. 2	exp_16	exp
F: capacity, both axes	§8	Ch. 11 §7	Tab. 3, Fig. 7, Fig. 6	exp_16	exp
F: sampler convergence	§8	Ch. 9	Fig. 6	exp_16 MODE=calibrate	exp
learned innovation law	§7	Ch. 10	Fig. 5	exp_18 -parts density	exp